PREDICTIVE ANALYTIC PRINCIPLES · ACTL4003 Project

Predicting *Mesothelioma* Survival Using Gene Expression & DNA Methylation

A genomic classification study using TCGA data

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RNA-SEQ · DNA METHYLATION

ELASTIC NET · RANDOM FOREST

N = 86 MPM TUMOURS

THE CHALLENGE

Malignant Pleural Mesothelioma: rare, fatal — and difficult to predict

DISEASE CONTEXT

MPM is a **rare thoracic cancer** primarily caused by asbestos exposure, with highly variable patient outcomes not explained by histology alone.

THE GAP

Conventional clinical groupings only **partially explain** observed variation. Additional molecular factors likely influence prognosis (Hmeljak et al., 2018).

OUR GOAL

Use **multiomic genomic data** — RNA-Seq and DNA methylation — to improve individual-level 500-day survival prediction.

Better prognostic models allow more accurate identification of **high-risk patients** who benefit from targeted treatment, and **low-risk patients** who can avoid unnecessary intervention — improving both outcomes and cost-effectiveness.

Prior Work & Methodological Foundations

🔬 MESOTHELIOMA BIOLOGY

Hmeljak et al. (2018) — Identified molecularly defined MPM subgroups independent of histological classification via TCGA integrative profiling.

Mangiante et al. (2023) — Classified MPM via four axes: ploidy, morphology, adaptive immune response, and CpG methylator profile.

Di Genova et al. (2023) — Demonstrated multiomic integration better captures tumour heterogeneity than any single data type alone.

📐 STATISTICAL METHODS

Penalised Regression — Friedman et al. (2010) — reduces overfitting through shrinkage in high-dimensional settings with correlated predictors.

Random Forest — Liaw & Wiener (2002) — captures nonlinear effects and interactions without pre-specification; attractive for complex biology.

Yuan et al. (2023) — Methylation-based cancer prediction supports statistical learning in small-n, high-dimensional genomic settings.

Dataset Overview

86
MPM TUMOURS

85
WITH RNA-SEQ

86
WITH METHYLATION

500 days

Binary survival threshold

Response variable defined from survival information as whether a patient lived longer than 500 days.

METHYLATION DATA

Array-based DNA methylation profiles measuring epigenetic modification at CpG sites, reflecting regulatory state beyond gene sequence.

EXPRESSION DATA

RNA sequencing measuring gene activity across the transcriptome, capturing active biological processes in each tumour sample.

Model Framework

Multiomics view of the 500-day survival prediction pipeline

TCGA Source

Public mesothelioma cohort with matched clinical, expression, and methylation data

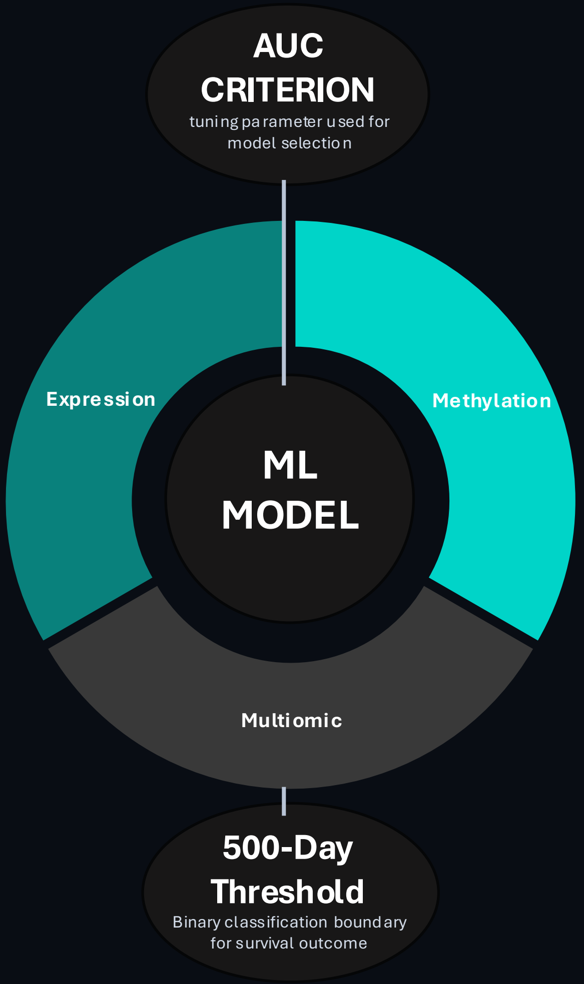
Cohort size (n=86)

Small-sample setting motivating careful validation and feature filtering

Biological Heterogeneity

Marked intertumoral variation motivates multi-omics modelling rather than reliance on morphology alone.

EXTERNAL



INTERNAL

Feature Filtering

Variance-based filtering and preprocessing reduce the omics layers to model-ready feature sets

Model Selection

Elastic net and random forest are compared across expression, methylation, and multiomic inputs

CV Strategy

Repeated cross-validation provides internal validation and tuning in a small cohort

Sensitivity / Specificity

Performance is interpreted through discrimination and clinically meaningful classification trade-offs

Variable Importance

Top genes and probes help explain which molecular features drive predictive signal

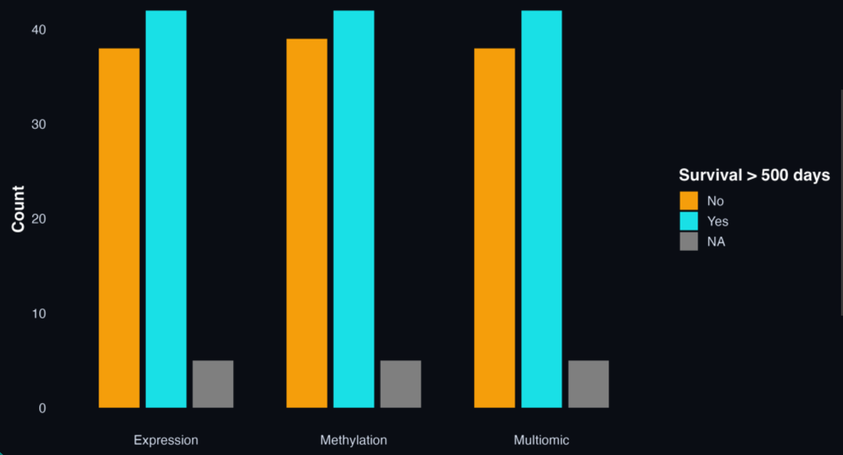
Data Construction

EDA takeaway

- Clinical missingness needed to be checked, but modelling focused on omics rather than expanded clinical benchmarking.
- PCA on top-variance features suggested substantial heterogeneity rather than clean class separation.
- This supported supervised modelling rather than relying on visual clustering.

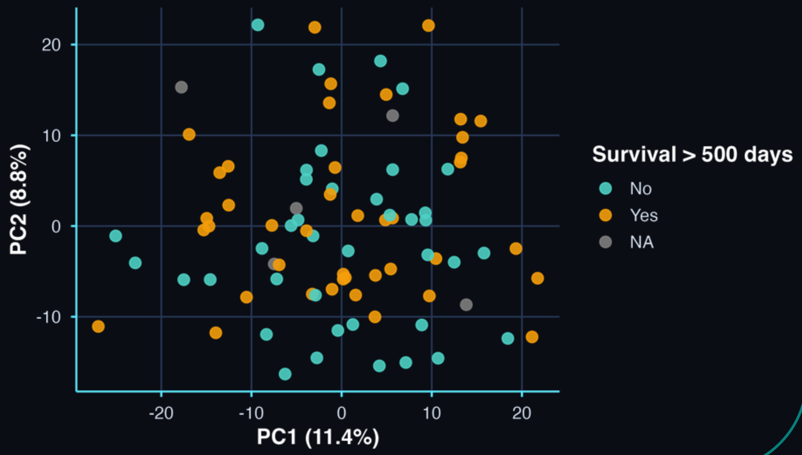
Dataset	Cases
Clinical master	86
Expression matched	85
Methylation matched	86
Multiomic matched	85

500-day Survival Class Balance by Cohort



Methylation PCA

Top-variance methylation features coloured by 500-day survival outcome

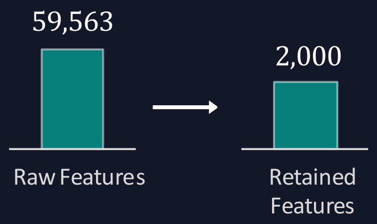


Data Construction

Master File n=86 → Expression Matched n=85 → Methylation matched n=86 → Multiomic matched n=85

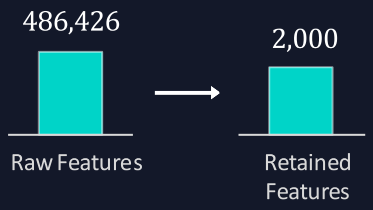
RNA EXPRESSION

- **Raw Features:** 59,563 genes
- **Labelled cases:** 80
- **Filtering:** complete genes + remove zero variance + top 2,000 by variance
- **Final Matrix:** 80 x 2,000



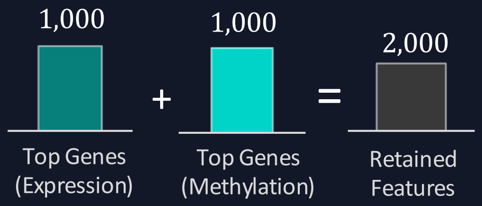
DNA METHYLATION

- **Raw Features:** 486,426 probes
- **Labelled Cases:** 81
- **Filtering:** Remove probes with >10% missingness; undefined variance; median impute; retain top 2,000 by variance.
- **Final Matrix:** 81 x 2,000



MULTIOMIC

- **Source:** combined expression + methylation
- **Labelled Cases:** 80
- **Construction:** top 1,000 expression +top 1,000 methylation
- **Total Retained Features:** 2,000
- **Final Matrix**: 80 x 2,000



FEATURE-SET LOGIC

Compact clinical cohort → matched omics cohorts → variance / missingness filtering → model-ready expression, methylation, and multiomic matrices

Two Algorithms · Three Predictor Sets

📉 ELASTIC NET

Penalised Logistic Regression

— Blends L1 (Lasso) and L2 (Ridge) penalties for flexible shrinkage

— Well-suited for correlated predictors in high-dimensional settings

— Implicitly performs feature selection or diffuse shrinkage

— Efficient linear benchmark for genomic classification

🌲 RANDOM FOREST

Nonlinear Ensemble Classifier

— Captures nonlinear effects and interactions without pre-specification

— Robust to irrelevant features in high-dimensional (p >> n) settings

— Produces variable importance scores linking predictions to biology

— Flexible nonlinear comparator for complex genomic applications

VALIDATION STRATEGY

5-fold cross-validation repeated 5 times — selected over a fixed train/test split because the small cohort (<86 patients) would make single-split estimates unstable. Tuning parameters selected by maximising cross-validated AUC (ROC). Class probabilities retained throughout.

Model Comparison Across Omics Settings

AUC COMPARISON	AUC	SENSITIVITY	SPECIFICITY
Multiomic Random Forest ★	0.756	0.629	0.748
Methylation Random Forest ★	0.747	0.644	0.720
Methylation Elastic Net	0.731	0.654	0.735
Multiomic Elastic Net	0.729	0.656	0.641
Expression Random Forest	0.722	0.630	0.696
Expression Elastic Net	0.717	0.556	0.753

★ Best overall: Multiomic RF achieved the highest AUC = 0.756; methylation RF was the strongest single-omics model with AUC = 0.747.

Sensitivity analysis

01 Methylation RF performance remained in the low-to-mid 0.70 AUC range across feature counts and cross-validation designs.

02 A smaller methylation feature set may reduce noise and improve stability.

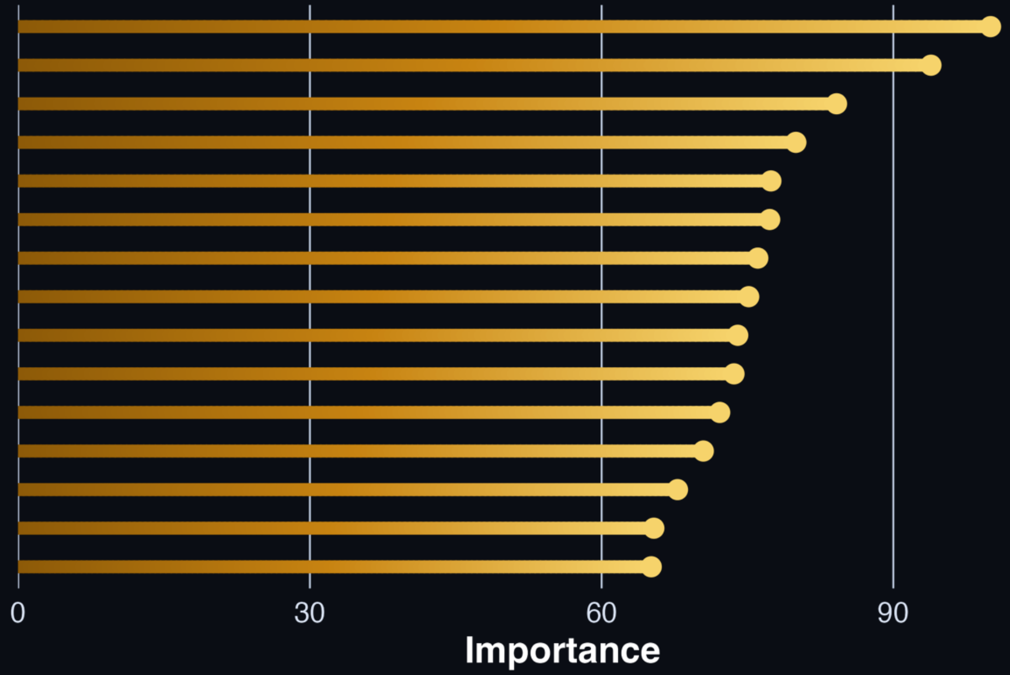
Interpretation

Performance was not driven by one arbitrary feature-count choice, although smaller feature sets sometimes improved stability.

SCENARIO	AUC	NOTE
1,500 probes	0.755	Highest sensitivity run
2,000 probes	0.749	Final tuned methylation RF
2,500 probes	0.741	Slightly lower
3×5 CV	0.734	Alternative resampling
10×3 CV	0.728	Alternative resampling

Top Methylation Features Mapped

Random Forest Variable Importance



MTMR7
Gene body

LAG3
TSS200 - enhancer

SPRED2
5′ UTR / enhancer

ZNF154
TSS1500

Several high-importance probes mapped to promoter-proximal or gene-associated regions.

Supporting the biological plausibility of the methylation signal.

Three Central Conclusions

01

METHYLATION IS THE STRONGEST SINGLE-OMICS SOURCE

- DNA methylation provided the strongest standalone prognostic signal
- The methylation-only random forest achieved **AUC = 0.747**
- Aligns with prior literature identifying CpG methylation as a key axis of MPM tumour heterogeneity

02

RANDOM FOREST OUTPERFORMS ELASTIC NET

- Random forest models generally produced stronger discrimination than elastic net
- Suggests that nonlinear relationships and interactions may be important in the molecular survival signal

03

MULTIOMIC INTEGRATION IMPROVES MODESTLY

- The best overall model was the **multiomic random forest**, with AUC = 0.756
- Improvement over methylation alone was modest
- Suggest that methylation captured much of the dominant prognostic signal

What Do These Results Actually Tell Us?

SENSITIVITY VS. SPECIFICITY TRADE-OFF

- Best models: Sensitivity ~0.63–0.64, Specificity ~0.72–0.75.
- Better at correctly identifying longer-surviving patients

- A meaningful fraction of high-risk patients would still be missed
- Models support coarse prognostic stratification, not standalone clinical decisions.

ELASTIC NET TUNING PATTERNS

- Expression EN → Lasso-leaning: a sparse gene subset drives prediction (sharp but noisy signals).

- Methylation EN → Ridge-like: signal spread across many correlated probes, reflecting broader regional methylation structure. Biologically plausible differences in data architecture.

PERFORMANCE CONTEXT

- AUC ~0.75 represents **moderate discrimination**

- Is meaningful predictive signal rather than noise.

- Only ~80 labelled patients

VARIABLE IMPORTANCE

- **Indicators of predictive usefulness**, not established causal drivers.

- Provide a starting point for further investigation nto mesothelioma progression pathways.

Limitations of This Analysis

SMALL SAMPLE SIZE

- ~80 labelled patients
- High-dimensional genomic setting
- Limits stability and makes estimates sample-sensitive

BINARY OUTCOME

- Compresses a more complex time-to-event process into a simpler classification task
- Discards information about the timing and censoring structure of survival data.

CONCATENATION STRATEGY

- Multiomic integration used direct feature concatenation.
- The limited gain should not be taken as evidence that more sophisticated integrative approaches (e.g. late-fusion, latent factor models) would also fail.

INTERNAL VALIDATION ONLY

- Results rely on repeated internal cross-validation
- External validation on an independent mesothelioma cohort is necessary for a stronger assessment of generalisability.

Directions for Future Research

PRIORITY 01

External Validation

Independent mesothelioma cohort to assess true generalisability beyond internal cross-validation.

PRIORITY 02

Advanced Integration

Late-fusion or latent-factor methods to better recover complementary information across omic layers.

PRIORITY 03

Survival Modelling

Move from binary classification to full time-to-event analysis using censored survival models.

PRIORITY 04

Biological Follow-up

Investigate high-importance methylation probes for mechanistic links to MPM progression pathways.

Genomic data can predict mesothelioma survival — *methylation leads the way*

FINDING 01

Multiomic RF Performs Best

Best overall model: AUC = 0.756. Combining expression and methylation produced the strongest discrimination, although the improvement over methylation alone was modest.

FINDING 02

Methylation Leads Single-Omics

Methylation RF achieved AUC = 0.747, outperforming expression-based models. Epigenetic variation appears to carry substantial prognostic signal.

FINDING 03

Random Forest Wins

Random Forest outperformed Elastic Net across the main omics settings, suggesting nonlinear flexibility helped capture molecular structure.

These findings show that omics-based survival prediction is feasible in mesothelioma, but the small cohort and high-dimensional feature space mean results should be treated as promising internal evidence rather than clinically validated prediction.

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2026